

The Journey

Programme Retrospective · Field in 2026 · Career Paths · Capstone Kickoff

You came in knowing how to prompt. You leave knowing how to build, deploy, and distribute AI systems.

Lecture Roadmap

A

The Journey — Module 1 → 10 Retrospective

20 min

B

Where the Field Stands in June 2026

15 min

C

Career Paths — How 10 Modules Map to Real Roles

12 min

D

Future Frontiers — Open Problems in Agentic & GenAI

13 min

E

Capstone Kickoff — Requirements, Deliverables, Timeline

18 min

F

Send-Off

12 min

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SECTION A

The Journey

Module 1 → 10

Where You Started — February 7, 2026

Week 1, Day 1: the Five-Level Framework set the map

L1 — Use

Interact with existing GenAI tools via APIs

L2 — Adapt

Prompt engineer, fine-tune, use RAG

L3 — Build

Construct agents, RAG pipelines, multimodal apps

L4 — Integrate

*Deploy, monitor, scale production
AI systems*

L5 — Innovate

*Pre-train, federate,
research novel
architectures*

"You entered at L1. The programme was designed to carry you to L4–L5 systematically."

M1

Practical GenAI Foundations

Weeks 1–2



Key Unlock: GenAI is not magic — it is token prediction over learned probability distributions. The Five-Level Framework gave you a compass for the entire journey.



Tools & Frameworks

- OpenAI / Anthropic APIs
- HuggingFace Hub
- Python environment



Mini-Project 1

Use-case analysis:

identified a real-world GenAI

opportunity and built the business case

M2

Prompt Engineering

Weeks 3–4

 **Key Unlock:** Structured prompting — zero-shot, CoT, tree-of-thought — transforms a raw model into a reliable reasoning tool without touching its weights.



Tools & Frameworks

- LangChain prompt templates
- OpenAI API
- DSPy (intro)



Mini-Project 2

Prompt portfolio:

10 production-grade templates

covering 5 task categories

M3

Large Language Models

Weeks 5–6



Key Unlock:

Understanding self-attention, tokenisation, and sampling parameters (temperature, top-p, top-k) means you can tune model behaviour — not just describe it.



Tools & Frameworks

- Transformer architecture
- HuggingFace Transformers
- API inference tuning



Mini-Project 3

API optimisation study:

benchmarked latency vs quality

across parameter configurations

M4

Multimodal AI Systems

Weeks 7–8

Key Unlock: Vision-Language Models (CLIP, ViT) and audio models (Whisper, TTS) revealed that language is only one modality — GenAI is inherently cross-modal.



Tools & Frameworks

- CLIP / ViT
- Whisper ASR
- PIL / OpenCV
- HuggingFace Audio



Mini-Project 4

Multimodal system:

built an image + text pipeline

that grounds LLM answers in visuals

M5

Introduction to Agentic AI

Weeks 9–10



Key Unlock: ReAct, MCP, RAG, and multi-agent coordination are the foundation of every production AI product being built today. This was the course's centre of gravity.



Tools & Frameworks

- LangChain / LangGraph
- MCP (Model Context Protocol)
- Pinecone / Qdrant
- A2A Protocol



Mini-Project 5

RAG Agent:

complete ReAct agent with

vector retrieval + tool use + traces

M6

Scientific ML & PINNs

Weeks 11–12



Key Unlock: Physics-Informed Neural Networks encode conservation laws directly into the loss function — the model cannot ignore physics even without labelled data. This is AI serving science.



Tools & Frameworks

- PyTorch
- DeepXDE
- Fourier Neural Operators
- SciPy / NumPy



Mini-Project 6

PINN application:

solved a 2D heat equation with

Physics-Informed loss — no simulation needed

M7

Foundational Model Fine-tuning

Weeks 13–14



Key Unlock:

LoRA, QLoRA, SFT, and DPO allow you to specialise any foundation model for a domain without retraining from scratch. RLHF connects model outputs to human preference.



Tools & Frameworks

- HuggingFace PEFT
- TRL (SFT/DPO/PPO)
- Weights & Biases
- QLoRA / bitsandbytes



Mini-Project 7

Domain fine-tune:

custom <1B model achieving

>20% improvement on domain benchmark

M8

AI Safety, Evaluation & Guardrails

Weeks 15–16



Key Unlock:

Evaluation is not a single metric — it is a multi-layered framework. Constitutional AI and guardrails show that safety is an engineering discipline, not an afterthought.



Tools & Frameworks

- OpenAI Evals
- RAGAS
- NeMo Guardrails
- Anthropic Constitutional AI



Mini-Project 8

Safety evaluation suite:

comprehensive evaluation framework

covering hallucination, bias, and toxicity

M9

LLMOps — Pre-training, Serving & Production

Weeks 17–19**Key Unlock:**

vLLM, PagedAttention, FlashAttention, Chinchilla scaling laws, and MCP-in-Production are the engineering layer that makes agents run reliably at scale — the largest module for a reason.

**Tools & Frameworks**

- vLLM / PagedAttention
- Docker / Kubernetes
- FastAPI
- Prometheus / Grafana
- FastMCP

**Mini-Project 9**

Production agent deployment:

containerised LangGraph agent

with full monitoring and CI/CD pipeline

M10

Federated & Distributed AI Learning

Weeks 20–21



Key Unlock:

Data parallelism, Ring-AllReduce, FedAvg, ZeRO, and federated privacy-preserving learning are the infrastructure of AI at enterprise and societal scale.



Tools & Frameworks

- PyTorch DDP / FSDP
- Flower FL framework
- NVIDIA FLARE
- HuggingFace Accelerate



Mini-Project 10

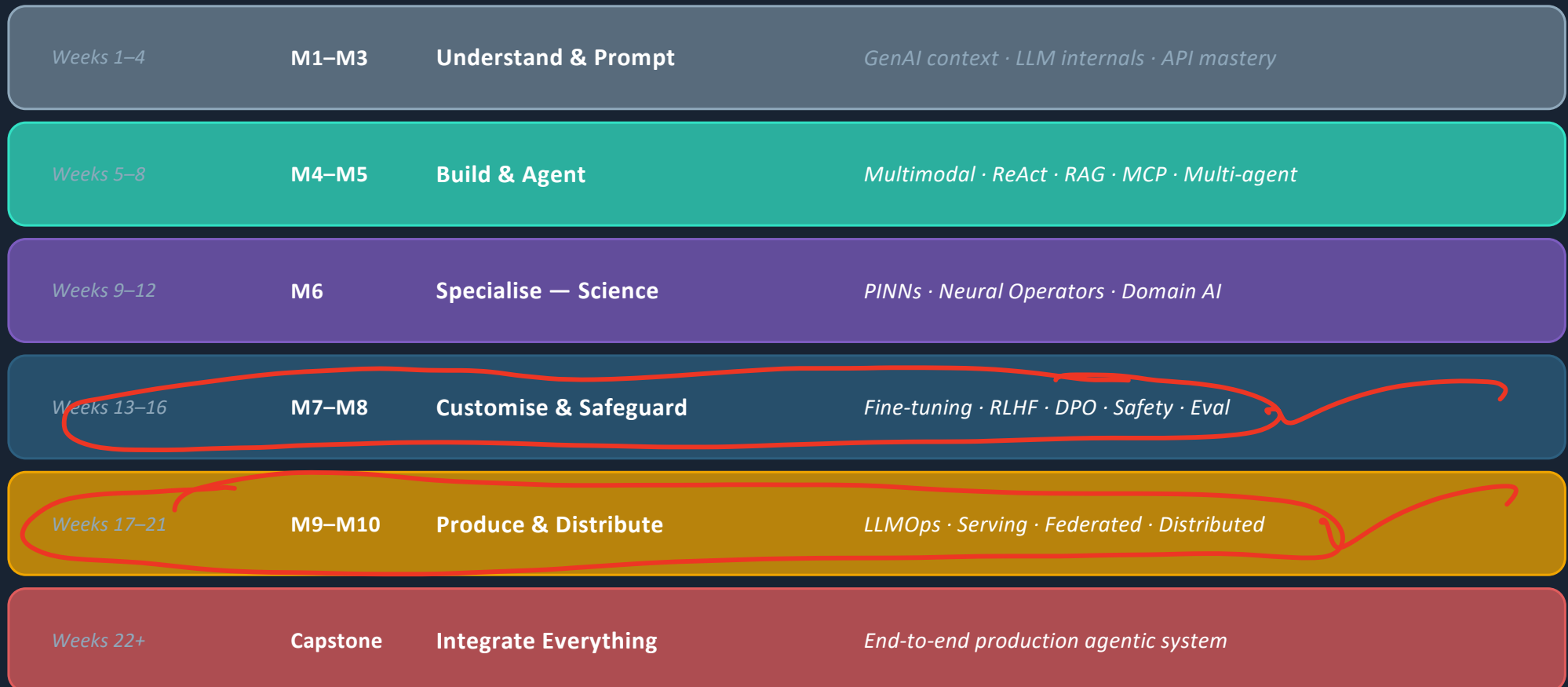
Federated learning system:

multi-client FL pipeline

with differential privacy and SecAgg

The Progressive Skill Stack — Built Layer by Layer

Each module added a measurable, deployable capability



Your Portfolio — 10 Projects Built

A GitHub record of progression from using to innovating



MP1: GenAI use-case analysis & business case

M1 — Foundations



MP2: Prompt engineering portfolio (10 templates)

M2 — Prompting



MP3: API parameter optimisation study

M3 — LLMs



MP4: Multimodal image+text grounding pipeline

M4 — Multimodal



MP5: RAG agent with ReAct + vector retrieval

M5 — Agentic AI



MP6: PINN solving 2D heat equation

M6 — Scientific ML



MP7: Domain-adapted fine-tuned <1B model

M7 — Fine-tuning



MP8: Safety evaluation suite

M8 — AI Safety



MP9: Containerised production agent + monitoring

M9 — LLM Ops



MP10: Federated learning system with DP+SecAgg

M10 — Federated AI

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SECTION B

Where the Field Stands in 2026

What practitioners face — not what research papers claim

The Field in June 2026 — Four Defining Shifts

What has changed since this programme began in February



Test-Time Compute Scaling

o1, o3, Gemini Thinking — reasoning at inference rather than training. Scale shifts from parameter count to compute-per-query.



Model Merging & Composition

SLERP, TIES, DARE allow specialists to be merged post-training. One team trains, many benefit — a new fine-tuning paradigm.



MCP as Infrastructure Standard

Model Context Protocol became the de-facto connector standard. Claude, Gemini, and GPT-4o all support it. You built on the winning protocol.



Sovereign & Edge AI

Governments and enterprises now demand on-premise, privacy-preserving inference. Federated learning (M10) and model compression (M10) are production requirements, not research exercises.

Every org will have their
own 22M (Fm)!

AI Platform

Hybrid (Frontier model)

Agentic AI in Production — June 2026



What real systems look like — and which course modules built each layer

User / Business Logic	REST / gRPC interface · streaming responses · rate limiting	M9 — LLMOps
Agent Orchestrator	LangGraph state machine · tool registry (MCP) · human-in-the-loop gates	M5 + M9
Reasoning Layer	LLM backbone (fine-tuned) · ReAct loop · self-reflection	M3 + M7 + M5
Memory & Retrieval	Vector DB (Qdrant) · RAG pipeline · semantic search	M5 — RAG
Safety & Eval	Guardrails · Constitutional AI · hallucination checks · eval harness	M8
Infrastructure	Docker · vLLM · Prometheus · CI/CD · model versioning	M9 — LLMOps

The Last-Mile Problem

Most AI projects that work in research fail in production — and you now know why

Industry reality

~85% of ML projects never reach production (Gartner 2024). The failures are not in model accuracy — they are in serving latency, cost, safety, observability, and maintenance.

LLMops

Data drift

Model performance degrades silently as input distributions shift over time

Fix: M9 monitoring

Safety gaps

Guardrails designed on test sets fail on adversarial real-world prompts

Fix: M8 evaluation suite

Latency & cost

p99 latency and per-query cost are not measured until production — too late

Fix: M9 vLLM / quantisation

Maintenance

No CI/CD for models means fixes take weeks and rollbacks are manual

Fix: M9 CI/CD pipeline

SECTION C

Career Paths

How the 10 modules map to real roles in industry



10 Modules → 5 Career Roles

Your strongest modules determine your natural path

Role	Modules	What You Build	Key Tools
AI / ML Engineer	M3,M5,M7,M9	<u>LLM APIs, agent backends, production services</u>	vLLM, <u>FastAPI</u> , <u>LangGraph</u>
Agentic Systems Engineer	M5,M8,M9	Multi-agent, <u>MCP integrations</u> , HITL	LangGraph, MCP, <u>Evals</u>
MLOps / AI Platform	M7,M8,M9,M10	Serving, CI/CD, monitoring, distributed	vLLM, K8s, Prometheus, FSDP <i>DevOps</i>
AI Safety / Evaluation	M2,M7,M8	Eval harnesses, guardrails, red-teaming	RAGAS, TRL, CAI, Evals <i>CoS</i>
Applied / Research Scientist	M3,M6,M7,M10	<u>Domain models</u> , PINNs, federated, papers	PyTorch, DeepXDE, Flower

No role is wrong — your capstone specialisation is the first data point in your career narrative.

What Employers Look for in 2026



Three signal types — and where this programme built each one

1

Can you build and deploy?

A working production system is worth 10 well-described skills. Your MP9 containerised agent and capstone are the evidence. ~~GitHub links matter.~~

MP9, Capstone

2

Do you understand the fundamentals?

Self-attention, LoRA rank, FedAvg convergence, constitutional AI. Interviewers test depth not breadth. Modules 3, 7, 8, 10 gave you the depth.

M3, M7, M8, M10

3

Can you evaluate and safeguard?

Every team is burned by hallucinations and safety failures. Evaluation fluency (RAGAS, LLM-as-judge, red-teaming) is the fastest-growing hiring signal.

M8, MP8

4

Can you reason about trade-offs?

Quantisation vs accuracy, LoRA rank vs capacity, FL communication vs convergence. Knowing when not to use a hammer distinguishes engineers.

All modules

5

Do you know the agentic stack?

ReAct, MCP, LangGraph, multi-agent patterns, human-in-the-loop. This course's M5+M9 combination is rare — ~~most~~ programmes still teach only fine-tuning.

M5, M9

The Five Learning Paths — Where Did You Land?

Your mini-project pattern reveals your natural specialisation

Path
1

API-First Developer

14 wks · M1→M2→M3→M5→M9→M10

Build and integrate LLM-powered applications. Fast path to shipping.

Path
3

Agentic Systems Builder

16 wks · Deep focus M5, M9

Production agents, MCP integrations, multi-agent coordination. Highest demand in 2026.

Path
2

RAG Specialist

16 wks · Adds M4 deep dive

Design retrieval systems that ground LLMs in enterprise knowledge.

Path
4

Model Developer

18 wks · Adds M7, M8

Domain-adapted fine-tuned models, safety-aligned, with eval suites.

Path
5

Full-Stack AI Developer

27 wks · All 10 modules

End-to-end: prompt → fine-tune → deploy → federate → monitor. This is you.

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SECTION D

Future Frontiers

Open problems in Agentic & Generative AI — 2026 and beyond

Open Problems in Agentic AI

What the best engineers and researchers are actively working on



Long-horizon reliability

Agents fail silently over multi-day tasks. No production system reliably completes >50-step autonomous plans without intervention. Evaluation frameworks for long-horizon tasks don't exist yet.



Multi-agent trust & alignment

When agents delegate to sub-agents, alignment breaks: the principal's intent degrades across hops. There is no verified protocol for multi-agent value inheritance.



Tool use at scale

Function calling works for 10–20 tools. With 500 tools, selection accuracy collapses. Hierarchical tool registries and semantic routing are active research areas.



Memory architecture

Episodic, semantic, and procedural memory should behave differently. Current LLM-based memory systems conflate them. LangGraph state is a workaround, not a solution.



Evaluation for agentic systems

Single-turn benchmarks (MMLU, HumanEval) are irrelevant for agents. Trajectory-level evaluation, tool-use accuracy, and goal-completion rates have no standard.



Human-AI collaboration design

Human-in-the-loop should not mean 'human interruption-in-the-loop'. The interaction model for oversight without friction is an open UX and systems research problem.

Open Problems at the Model Level

The questions that will define the next generation of models

Hallucination root cause

We can detect hallucinations but cannot reliably prevent them. Is it data, architecture, or training objective? Active research in 2026 points to retrieval grounding and CoT verification as partial solutions.

Reasoning vs memorisation

Chain-of-thought improves outputs but it is unclear whether models reason or pattern-match. The distinction matters for safety guarantees on novel situations.

Continual learning

Models trained once become stale. Continual fine-tuning causes catastrophic forgetting. Elastic Weight Consolidation and LoRA-based CL are partial answers — the problem is unsolved.

Scaling laws beyond next-token

Chinchilla optimised parameter-data ratios for next-token prediction. Test-time compute scaling (o1/o3) suggests a new frontier. The governing laws are not yet understood.

Open Problems in FL & Distributed AI

From the lens of Agentic and GenAI — not just distributed systems



FL for LLMs (>1B)

FedPETuning (federated LoRA) works. Full-weight federated LLM training requires INT4 + Top-K sparsification that remains prohibitively expensive at production scale.



Agentic FL

Autonomous agents learning federatedly face non-stationarity — each agent changes the environment, breaking the i.i.d. assumption of FedAvg. No algorithm solves this yet.



Byzantine robustness at scale

Krum and median aggregation work when <30% of clients are malicious. Beyond that threshold, provable guarantees break. Practical production FL ignores this.



Incentive mechanism design

Shapley value attribution is the theoretically correct incentive signal but costs $O(2^N)$ to compute. All practical approximations have exploitable gaps.



Non-IID convergence guarantees

SCAFFOLD and FedNova improve empirical convergence. Theoretical convergence guarantees on arbitrary non-IID distributions remain an open problem.



Sovereign AI + FL

Governments want data sovereignty (country-level silos) and collaborative global models simultaneously. The regulatory + technical design space is completely unexplored.

Where to Go Next

Concrete actions from here

1

Build in public

Publish your MP9 and capstone on GitHub. Write a 500-word blog post per project. The field rewards documented builders.

2

Follow the research

ArXiv cs.AI and cs.LG weekly. Key venues: NeurIPS, ICLR, ICML, ACL. RSS: Hugging Face Daily Papers, The Batch.

3

Go deep on one area

Breadth got you to L4. Depth takes you to L5. Pick one: agentic systems, safety, scientific ML, or distributed AI — and publish.

4

Contribute to open source

LangGraph, Flower FL, DeepXDE, TRL are all active. A merged PR is worth more than any certification on a CV.

5

Apply to the frontier

Anthropic, Google DeepMind, OpenAI, Microsoft Research, Cohere, Mistral, and IISc all hire from programmes like this. Start with your strongest project.

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SECTION E

Capstone Kickoff

4 weeks · 40+ hours · teams of 5–8 · July 26, 2026

What the Capstone Requires

Five non-negotiable deliverables — all five must be present

1

LLM / Model Component

A trained or fine-tuned model (<1B parameters). Documented model card. Performance benchmarks against baseline. Architecture choices justified.

2

Application Layer

Working application: RAG pipeline, API endpoints, agent interface, or domain tool. Must solve a real problem — not a tutorial demo.

3

Safety & Evaluation

Comprehensive eval metrics. Guardrails implemented. Hallucination and bias analysis. Red-team results documented. Limitations stated honestly.

4

Production Readiness

Docker container. Deployment infrastructure. Prometheus monitoring dashboard. CI/CD pipeline. Cost tracking. Scalability documented.

5

Documentation

Model card (HuggingFace format). Technical documentation. Deployment guide. User guide. Lessons learned — the hard ones count double.

How Every Module Powers the Capstone



Nothing you learned was orphaned — it all connects here

M1

Business case & use-case framing

Your problem statement and ROI justification

M3

LLM selection & API tuning

Choosing the right backbone and inference parameters

M5

Agent architecture

ReAct loop, MCP tool integration, multi-agent patterns

M7

Fine-tuned model

Your <1B model, LoRA-adapted, with documented training run

M9

Production infrastructure

vLLM serving, Docker, monitoring, CI/CD, MCP server

M2

Prompt engineering

System prompts and chain-of-thought for your agent

M4

Multimodal inputs (if applicable)

Vision or audio components in your application layer

M6

Domain knowledge (if STEM)

PINN or operator learning components for science capstones

M8

Safety & evaluation suite

Guardrails, red-team results, bias analysis, hallucination metrics

M10

Distributed training (optional)

FL pipeline for data-sensitive capstones

Capstone Timeline — 4 Weeks

June 27 → July 26, 2026 · Teams of 5–8

1

Week 1
Jun 27 – Jul 3

Planning & Architecture

Problem definition · solution design · technology selection · team roles · milestone planning · first commit to GitHub

✓ Frozen architecture document

2

Week 2
Jul 4 – Jul 10

Development

Data preparation · model training/fine-tuning · agent implementation · RAG pipeline · system integration · iteration

✓ Working prototype with all 5 components

3

Week 3
Jul 11 – Jul 17

Production Hardening

Safety implementation · eval framework · monitoring setup · optimisation · documentation · stress testing

✓ Deployed system with monitoring live

4

Week 4
Jul 18 – Jul 26

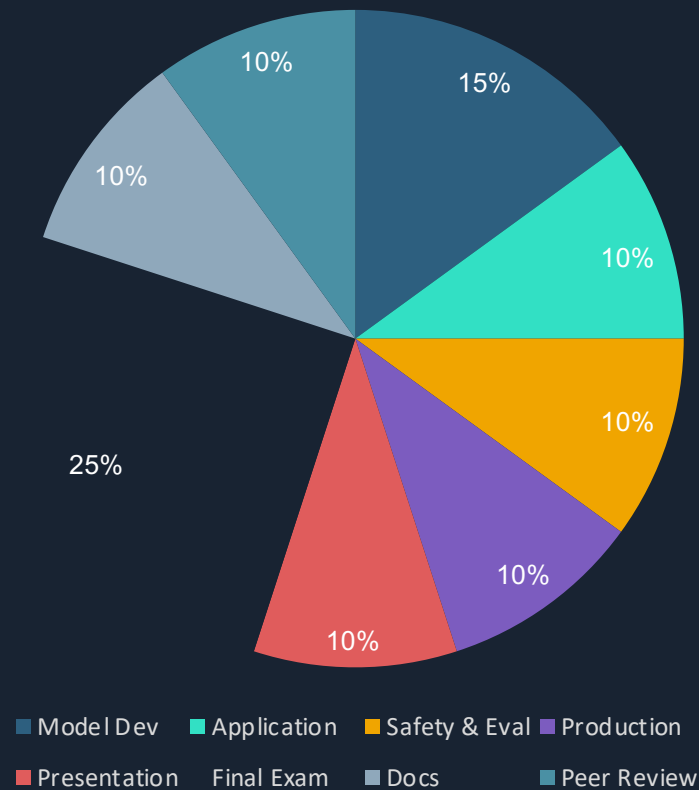
Finalisation & Presentation

Final testing · campus presentation prep · 15-20 min live demo · technical Q&A · peer feedback · submission

✓ Final submission + live demonstration

Capstone Grading Structure

Assessment is weighted toward demonstrated working systems, not documentation



Model Dev & Training	15%
<i>Architecture choices justified; bench mark vs baseline documented</i>	
Application Layer	10%
<i>Solves a real problem; demonstrable end-to-end</i>	
Safety & Evaluation	10%
<i>Guardrails live; red-team results honest</i>	
Production Readiness	10%
<i>Running deployment — not just a notebook</i>	
Presentation & Demo	10%
<i>Live system demo + Q&A defence</i>	
Final Exam (B5)	25%
<i>Create & Evaluate — open-ended design problem</i>	
Documentation	10%
<i>Model card + deployment guide + lessons learned</i>	
Peer Review	10%
<i>Quality of feedback given to other teams</i>	

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SECTION F

Send-Off

26 weeks · from prompting to production · what happens next

The Skill Stack You Leave With

Cumulative capability — each layer permanent

✓ **Use**

Converse fluently with frontier models. Craft precise prompts. Control output with temperature, top-p, top-k.

✓ **Adapt**

Fine-tune any open-weight model with LoRA/QLoRA. Use RLHF and DPO for alignment. Evaluate rigorously.

✓ **Build**

Construct ReAct agents. Design RAG pipelines. Integrate MCP. Orchestrate with LangGraph.

✓
Customise

Train domain-adapted <1B models. Apply safety guardrails. Red-team your own systems.

✓ **Innovate**

Deploy production LLM systems. Train federatedly. Apply PINNs to science. Scale with DDP/FSDP.

The Three Things That Will Actually Matter

After 26 weeks, here is what separates practitioners who grow from those who plateau

1

Build something real every month

The field moves every 90 days. Anyone who stops building loses the thread. Your capstone is the beginning, not the end. One deployed project per month keeps your instincts sharp.

2

Evaluate before you ship — always

Every hallucination that reaches a user erodes trust that takes months to rebuild. Build the eval harness first. Run it last. This is the single most-skipped step in industry.

3

Stay close to the research, but be suspicious of it

Read the papers. Reproduce the results. Most benchmarks are gamed. Real performance shows up when your agent tries to book a flight or answer a clinical question — not on MMLU.

What the Programme Actually Gave You



Beyond the skill stack

1

A mental model for the whole field

You can place any new technique — LoRA, RLHF, MCP, FedProx — on the map. That navigational fluency is rarer than any individual skill.

2

A cohort of peers

The people in this room are building AI systems in 2026. They are your future collaborators, co-founders, and referrals. The cohort outlasts the curriculum.

3

A portfolio of real work

10 mini-projects + a production capstone. Code that runs. Numbers that were measured. Systems that were deployed. Not slides about what you might build.

4

A foundation for research

You now understand the papers. You can run ablations. You can reproduce baselines. The door to publishing is open — if you choose to walk through it.

5

Credibility to say no

The most valuable professional skill is knowing when not to use AI, when a solution is over-engineered, and when a simpler approach is better. You now have the depth to judge.

Programme Complete



You Are Now Ready to Build

*"You came in knowing how to prompt.
You leave knowing how to build, fine-tune, deploy, evaluate, and distribute AI systems.
The field is yours."*

300+

Concepts Mastered

10

Mini-Projects Delivered

26 weeks

Jan → July 2026

1

Production System — Your Capstone